Exp. No: 4

Analysis of Continuous Charge Distribution in Electric Field (Line, Surface and Volume Charge Density)

Aim:

To write program for continuous Charge Distribution in electric field. The different types of charge density are (a) Linear Charge Density, (b) Surface Charge Density, (c) Volume Charge Density using MATLAB.

Objectives:

1. To compute line, surface and volume charge densities for a given continuous charge distribution.

2. To analyze how electric field intensity varies for each type of charge distribution.

3. To compare the computed charge densities against theoretical or analytical expression

Program:

Test case (i) :

Determination of line, surface and volume charge density for given charge, length and radius

```
clc;

clear;

q=input("Enter the charge value is:");

l=input("Enter the length value is:");

% Line charge density
```

```
rho_l = q/l;
disp('the line charge density is');
disp(rho_l);
% Surface charge density
r=input("Enter the radius value is:");
rho_s = q/(4*pi*r^2);
disp('the surface charge density is');
disp(rho_s);
% Volume charge density
rho_v = q/((4/3)*pi*r^3);
disp('the volume charge density is');
disp(rho_v);
% Plot
x = linspace(0.1,10,50);
y = rho_l./x;
z = rho_s./(x.^2);
v = rho_v./(x.^3);
figure;
subplot(1,3,1);
plot(x,y,'LineWidth',2);
title('Line charge density');
xlabel('Length');
subplot(1,3,2);
plot(x,z,'LineWidth',2);
title('Surface charge density');
```

```
xlabel('Radius');

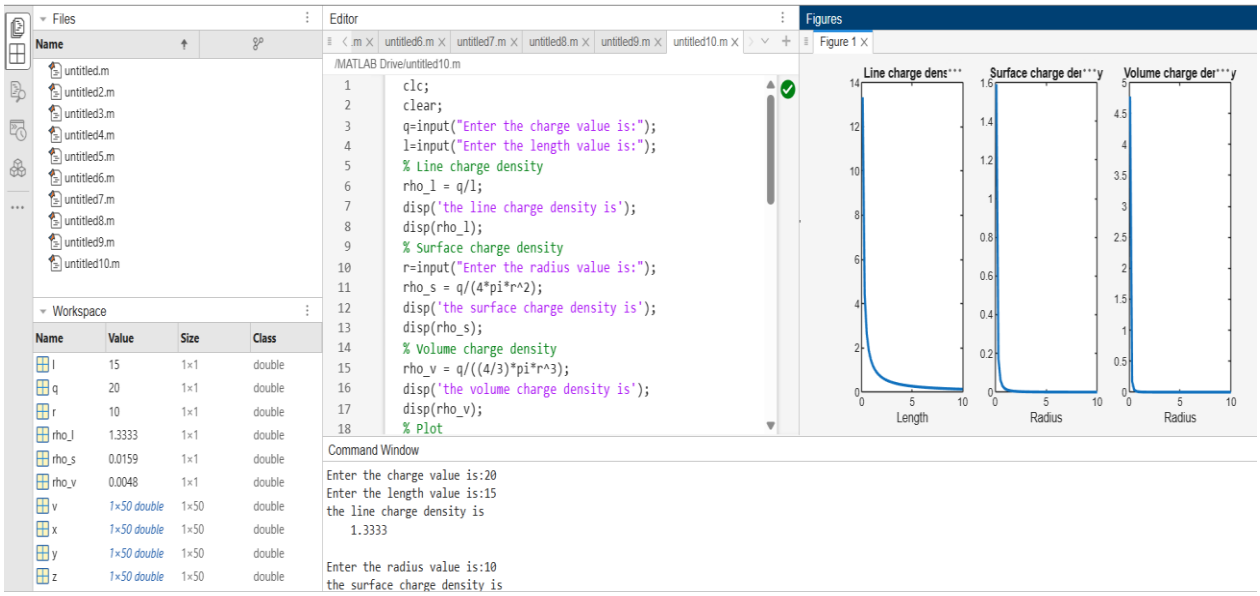
subplot(1,3,3);

plot(x,v,'LineWidth',2);

title('Volume charge density');

xlabel('Radius');
```

output:



Test case (ii) :

A long wire of length L = 10 m carries a uniform charge of Q = 50 μC. Calculate the line charge density and plot electric field variation

Program:

```
clc;

L = 10;

Q = 50e-6;

lambda = Q/L;

disp("Line charge density:");

disp(lambda);
```

```
r = linspace(0.1,5,100);

k = 9e9;

E = k*lambda./r;

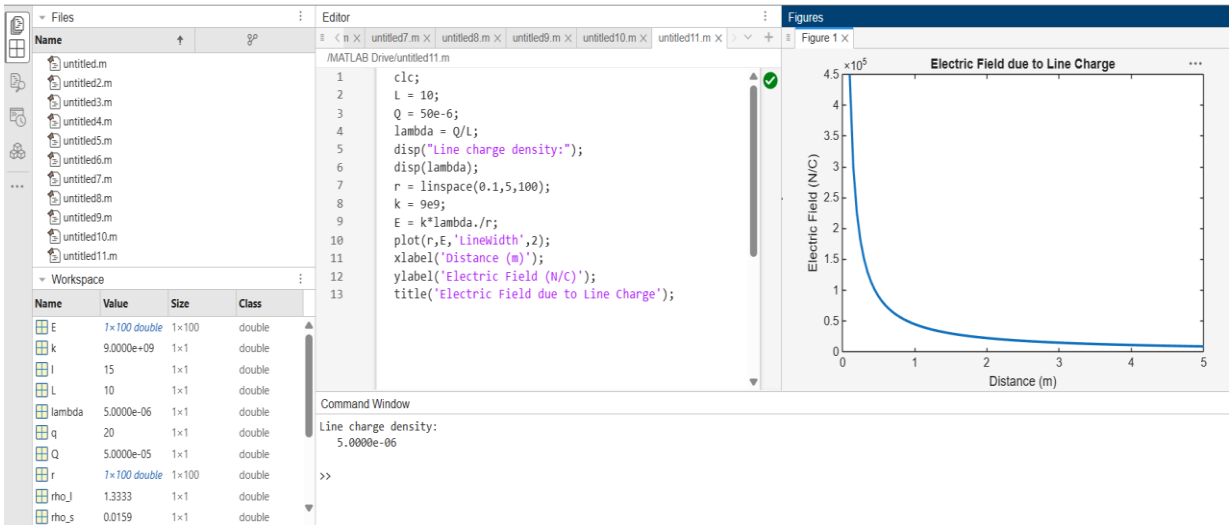
plot(r,E,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Electric Field (N/C)');

title('Electric Field due to Line Charge');
```

output:



Test case (iii) :

A uniformly charged disk of radius R = 0.5 m with total charge Q = 10 μC. Calculate surface charge density and electric field

Program:

```
clc;

R = 0.5;

Q = 10e-6;

A = pi*R^2;
```

```
sigma = Q/A;

disp("Surface charge density:");

disp(sigma);

z = linspace(0.1,2,100);

k = 9e9;

E = (k*sigma*R^2)./(2*(z.^2 + R^2).^(3/2));

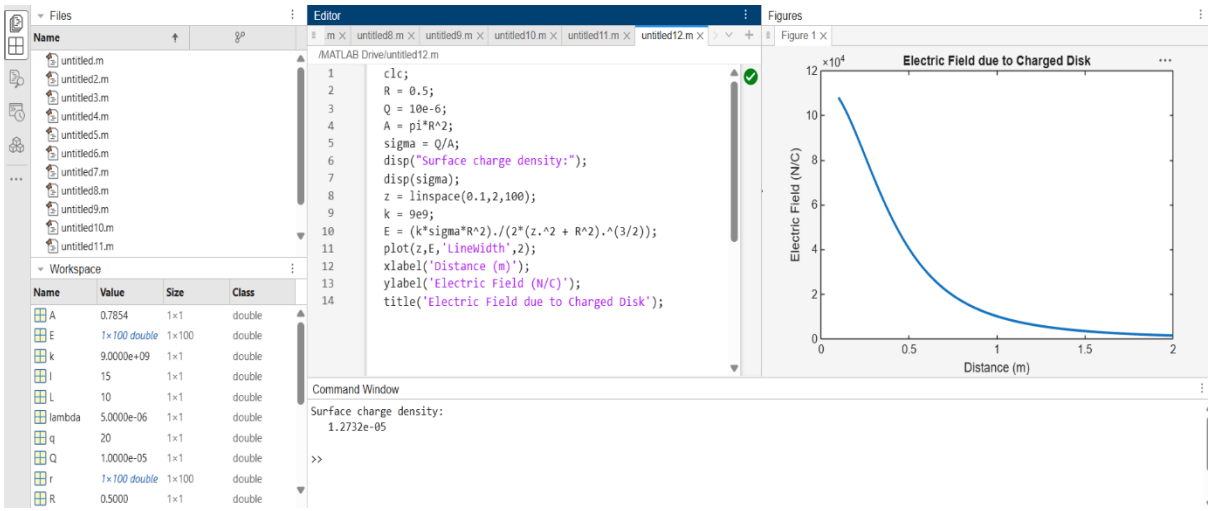
plot(z,E,'LineWidth',2);

xlabel('Distance (m)');

ylabel('Electric Field (N/C)');

title('Electric Field due to Charged Disk');
```

output:



Test case (iv)

A uniformly charged sphere of radius R = 2 m with total charge Q = 20 μC. Calculate volume charge density and electric field inside and outside 3

Program:

```
clc;

R = 2;

Q = 20e-6;
```

```
V = (4/3)*pi*R^3;

rho = Q/V;

disp("Volume charge density:");

disp(rho);

r = linspace(0.1,3,100);

epsilon0 = 8.854e-12;

k = 1/(4*pi*epsilon0);

E_inside = (rho.*r)/(3*epsilon0);

E_outside = k*Q./(r.^2);

plot(r,E_inside,'LineWidth',2); hold on;

plot(r,E_outside,'LineWidth',2);

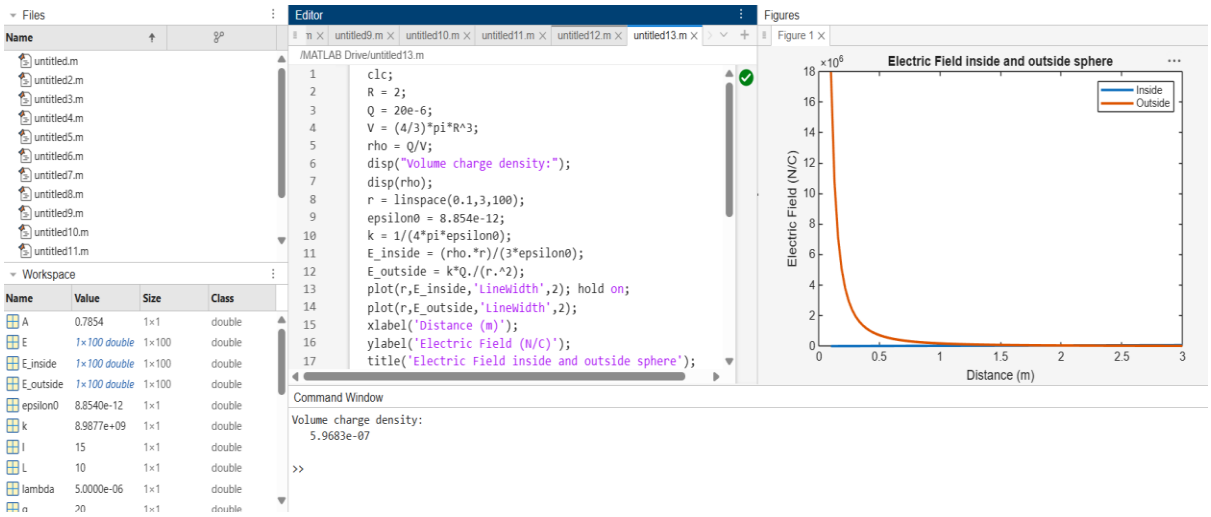
xlabel('Distance (m)');

ylabel('Electric Field (N/C)');

title('Electric Field inside and outside sphere');

legend('Inside','Outside');
```

output:



Rubrics:

S. No.	Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weighting (%)	Marks given	Marks obtained
1	Identify physical quantities and force models	Identify physical quantities and force models	Correctly identifies && charge/current, && separation, && medium and && force direction && for both cases.	Identifies most && quantities and models && correctly.	Needs guidance in selecting the correct && relation or direction.	Uses incorrect force models or && quantities.	25	2.5	
2	Calculate electric and magnetic forces	Calculate electric and magnetic forces	Implements && Coulomb-force && and parallel && current force && calculations && correctly with units.	Calculations are mostly correct && with minor errors.	Some calculations && are correct but require && assistance.	Force && calculations are && invalid or absent.	25	2.5	
3	Evaluate force versus separation	Evaluate force versus separation	Correctly varies && separation, && plots trends and && checks theoretical && distance && dependence.	Trend and plots are correct with && minor omissions.	Limited sweep or && incomplete trend && interpretation.	Does not && establish a valid && relationship.	30	3	
4	Explain and report results	Explain and report results	Explains && attraction/repulsion && and reports && assumptions, && units and && conclusions clearly.	Conclusion is correct with && minor documentation gaps.	Conclusion is && incomplete or && partly inaccurate.	Conclusion is missing or || incorrect.	20	2	
	Total						**100**	**10**	